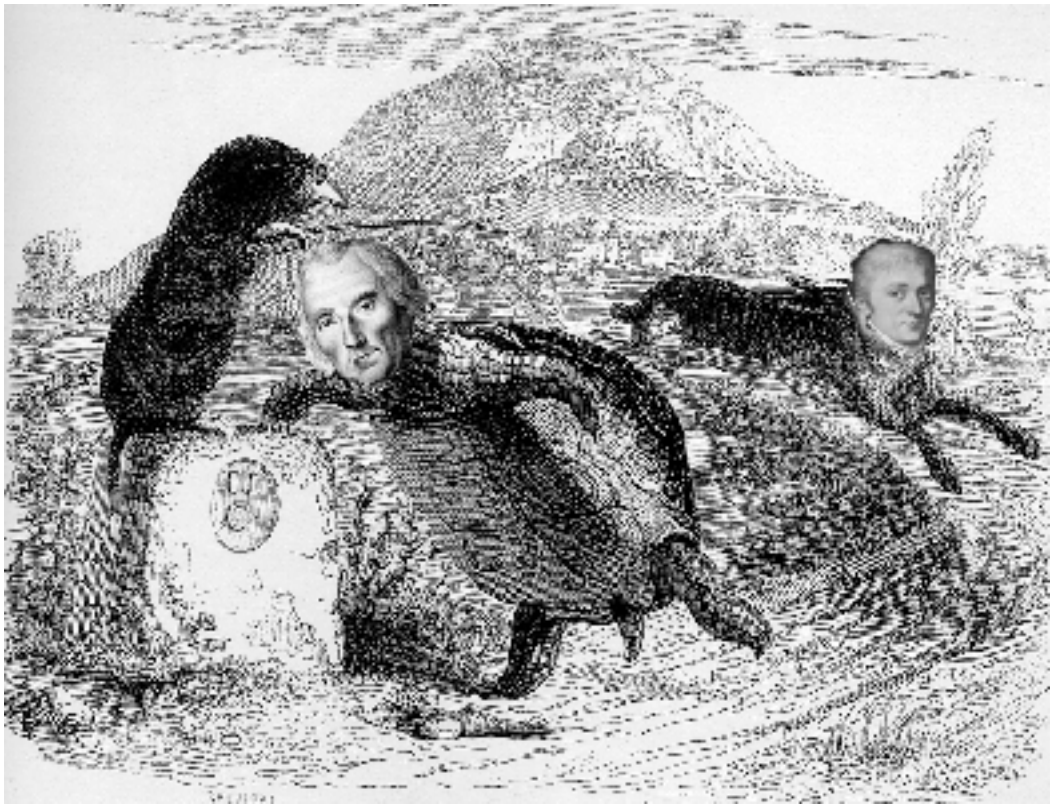
ℓ_1 TORTOISE GAINS ON ℓ_2 HARE

ROGER KOENKER



Tartu, Estonia: February 10, 1998. Archival sources announced here today the discovery of an unusual wood-etching/"photoprint" from the late 18th century depicting the conclusion of Stage 11 of Round G of the *Tour de Monde Statistique.* Previous *reportage* from this race by M. La Fontaine had suggested the possibility that the perennially over-confident Hare "Gauss" had been overtaken by the tortoise representing M. Laplace. The print, reproduced above, confirms quite conclusively, that the tortoise was leading at the stone marker ending stage G-11. However, attempts to obtain the official records of this stage of the race, presumably registered by the unidentified mole in sunglasses with the dowsing stick, have proven, thus far, unsuccessful.

1. An Historical Introduction

Since Gauss it has been generally accepted that ℓ_2 methods of combining observations by minimizing sums of squared residuals have significant computational advantages over earlier ℓ_1 methods based on minimizing sums of absolute residuals, advocated by Boscovich, Laplace and others. In 1887, six years after publishing his path breaking work in economics *Mathematical Psychics*, F.Y. Edgeworth began a series of papers "On a new method of reducing observations relating to several quantities." In fact the method was not *entirely* new. In the 1760's the Croatian Jesuit Roger Boscovich proposed estimating the ellipticity of the earth by solving a problem of the form:

$$\min \sum_{i=1}^{n} |y_i - \alpha - x_i'\beta|$$

subject to:

$$\sum_{i=1}^{n} (y_i - \alpha - x_i'\beta) = 0,$$

where y_i was the length of one degree of latitude measured at latitude θ_i and $x_i = \sin^2 \theta$. Somewhat later, Laplace showed that this problem could be solved by computing a weighted median, and provided an astonishingly complete theory of the limiting distribution of the resulting estimator $\hat{\beta}$, when $\alpha = 0$.

Edgeworth's new method, which he called the "plural median" was intended to revive the Boscovich/Laplace approach as a direct competitor to the least squares approach championed by Galton and others. Edgeworth (1888) proposed dropping the zero-mean constraint on residuals, arguing that it conflicted with the median intent of the absolute error approach. And appealing to the univariate results of Laplace, he conjectured that the plural median should be more accurate than the least squares estimator when the observations were more "discordant" than those from the Gausssian probability law. Finally, he proposed a rather arcane geometric algorithm for computing the plural median and remarked rather cryptically:

> ...the probable error is increased by about 20 percent when we substitute the Median for the Mean. On the other hand, the labour of extracting the former is rather less: especially, I should think in the case of many unknown variables. At the same time, that labour is more "skilled". There may be needed the attention of a mathematician; and, in the case of many unknowns, some power of hypergeometrical conception.

The "20 percent" is a bit optimistic. At the normal model the median would have confidence intervals which are about 25 percent wider than those based on the mean. But many of the details of Edgeworth's conjectures concerning the improvements achieveable by the plural median over comparable least squares methods of inference

in discordant situations have been filled in over the last 20 years. See, for example, Bassett and Koenker (1978), Koenker and Bassett (1982), Powell (1986), Gutenbrunner and Jurečková (1992), Gutenbrunner, Jurečková , Koenker and Portnoy (1993), and the three conference volumes of Dodge(1987,1992,1997).

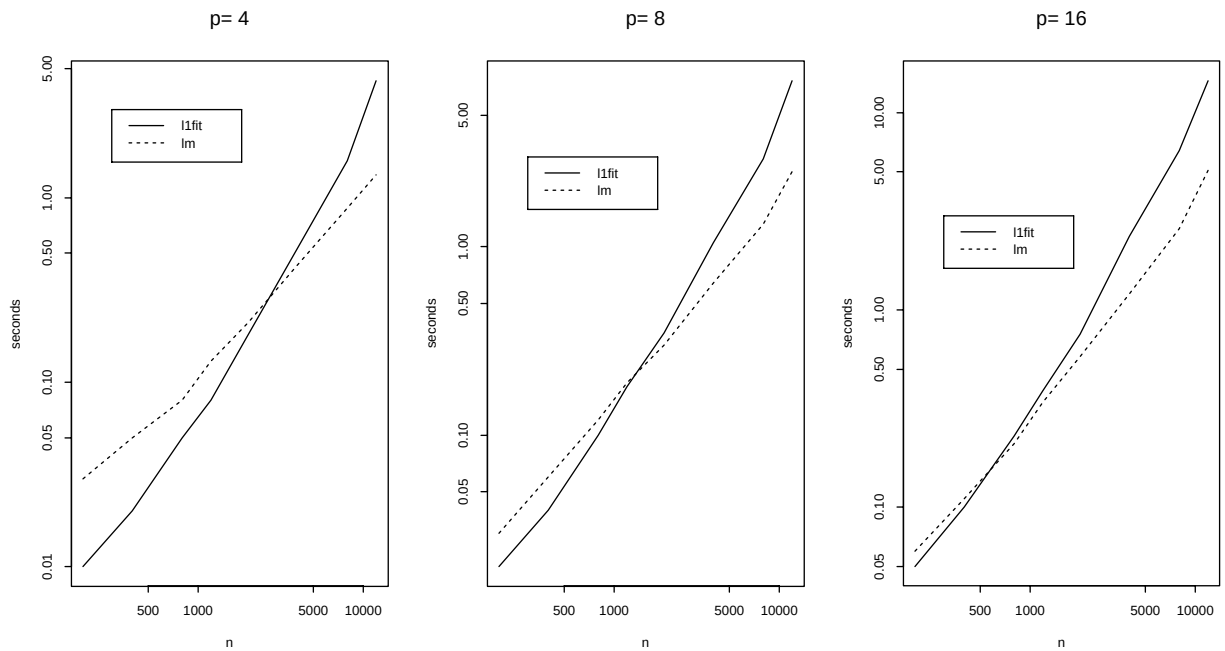


FIGURE 1.1. Timing comparison of ℓ_1 and ℓ_2 algorithms: Times are in seconds for the median of five replications for iid Gaussian data. The parametric dimension of the models is $p+1$ with p indicated above each plot, p columns are generated randomly and an intercept parameter is appended to the resulting design. Timings were made at 8 design points in n: 200, 400, 800, 1200, 2000, 4000, 8000, 12000. The solid line represents the results for the simplex-based Barrodale and Roberts algorithm, `l1fit(x,y)` in Splus, and the dotted line represents least squares timings based on `lm(y ~ x)`.

Even in the elementary one-sample setting we can see the merit in Edgeworth's claim: the median is less laborious to compute by hand, but perhaps requires more skill than finding the mean. In large samples, if we adopt the naive idea that finding the median requires a complete sorting of the n sample observations it is easy to reach the conclusion that the median needs $\mathcal{O}(n^2)$ operations while the mean can obviously be computed in $\mathcal{O}(n)$ operations. More careful sorting quite easily reduces $\mathcal{O}(n^2)$ to $\mathcal{O}(n \log n)$, but considerable "skill" was required to design the $\mathcal{O}(n)$ algorithm of

Floyd and Rivest (1975). Can something comparable be done to make ℓ_1 regression competitive with least squares in terms of computational complexity?

Edgeworth's conjectures about the computational complexity of ℓ_1 versus ℓ_2 regression methods were also partially vindicated by the development of the simplex method of linear programming by Dantzig and others in the late 1940's. Refinements of simplex designed for the ℓ_1 regression problem including the widely implemented Barrodale and Roberts (1973) algorithm proved quite competitive with least squares computational speed for problems up to a few thousand observations. In Figure 1.1 we illustrate this based on experience in Splus on a Sparc 20. Note that up to about 3000 observations for $p = 4$ parameters, or up to about n=1000, for $p = 8$, or $n = 300$ for $p = 16$, the Splus function `l1fit` embodying the algorithm of Barrodale and Roberts is actually faster than the QR decomposition algorithm for least squares embodied the the Splus function `lm()`. However in larger problems the simplex approach founders badly, exhibiting quadratic growth in cpu-time with n. By the time that we reach $n = 100,000$, with $p = 16$ for example, `l1fit` requires nearly an hour of Sparc 20 time while the equivalent least squares computation takes about 10 seconds.

2. Interior versus Exterior Methods

A forthcoming paper, Portnoy and Koenker (1997), explores two approaches which, taken together, provide some reason for optimism. I will briefly describe both approaches, relying on the full paper to fill in the details. Consider the median regression problem,

$$\min_{b \in \mathbb{R}^p} \sum_{i=1}^{n} |y_i - x_i' b| \tag{2.1}$$

which may be formulated as the linear program,

$$\min\{\ e'u + e'v \ \mid\ y = Xb + u - v, (u, v) \in \mathbb{R}_+^{2n}\}. \tag{2.2}$$

Note that we have simply decomposed the regression residual vector into its positive and negative parts, calling them u and v, and written the original problem as one of minimizing a linear function of the $2n$-vector (u, v) subject to n linear equality constraints and $2n$ linear inequality constraints. This "primal" linear program formulation of the ℓ_1-regression problem has an associated "dual" formulation in which we maximize with respect to a vector, $d \in \mathbb{R}^n$, which may be viewed as the vector of Lagrange multipliers associated with the equality constraints of the primal problem. This dual formulation is,

$$\max\{y'd \ \mid\ X'd = 0, \quad d \in [-1, 1]^n\}, \tag{2.3}$$

or equivalently, setting $a = d + \frac{1}{2} e_n$,

$$\max\{y'a \ \mid\ X'a = \tfrac{1}{2} X' e_n, \ \ a \in [0, 1]^n\}, \tag{2.4}$$

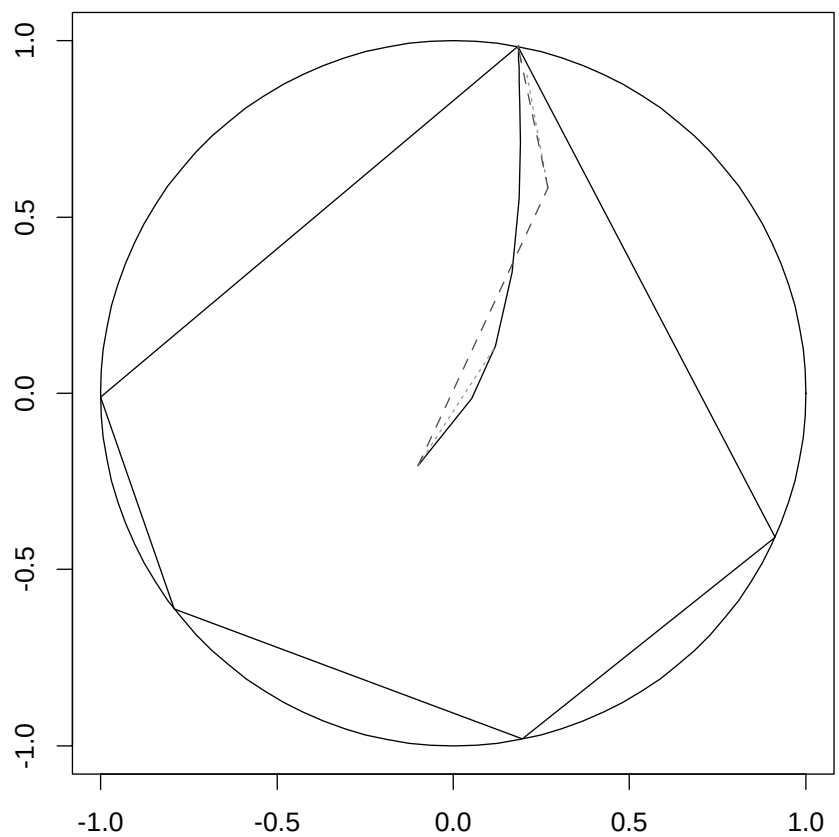


FIGURE 2.1. A Simple Example of Interior Point Methods for Linear Programming: The figure illustrates a random pentagon of which we would like to find the most northeast vertex. The central path beginning with an equal weighting of the 5 extreme points of the polygon is shown as the solid curved line. The dotted line emanating from the this center is the first affine scaling step. The dashed line is the modified Newton direction computed according to the proposal of Mehrotra (1992). Subsequent iterations are unfortunately obscured by the scale of the figure.

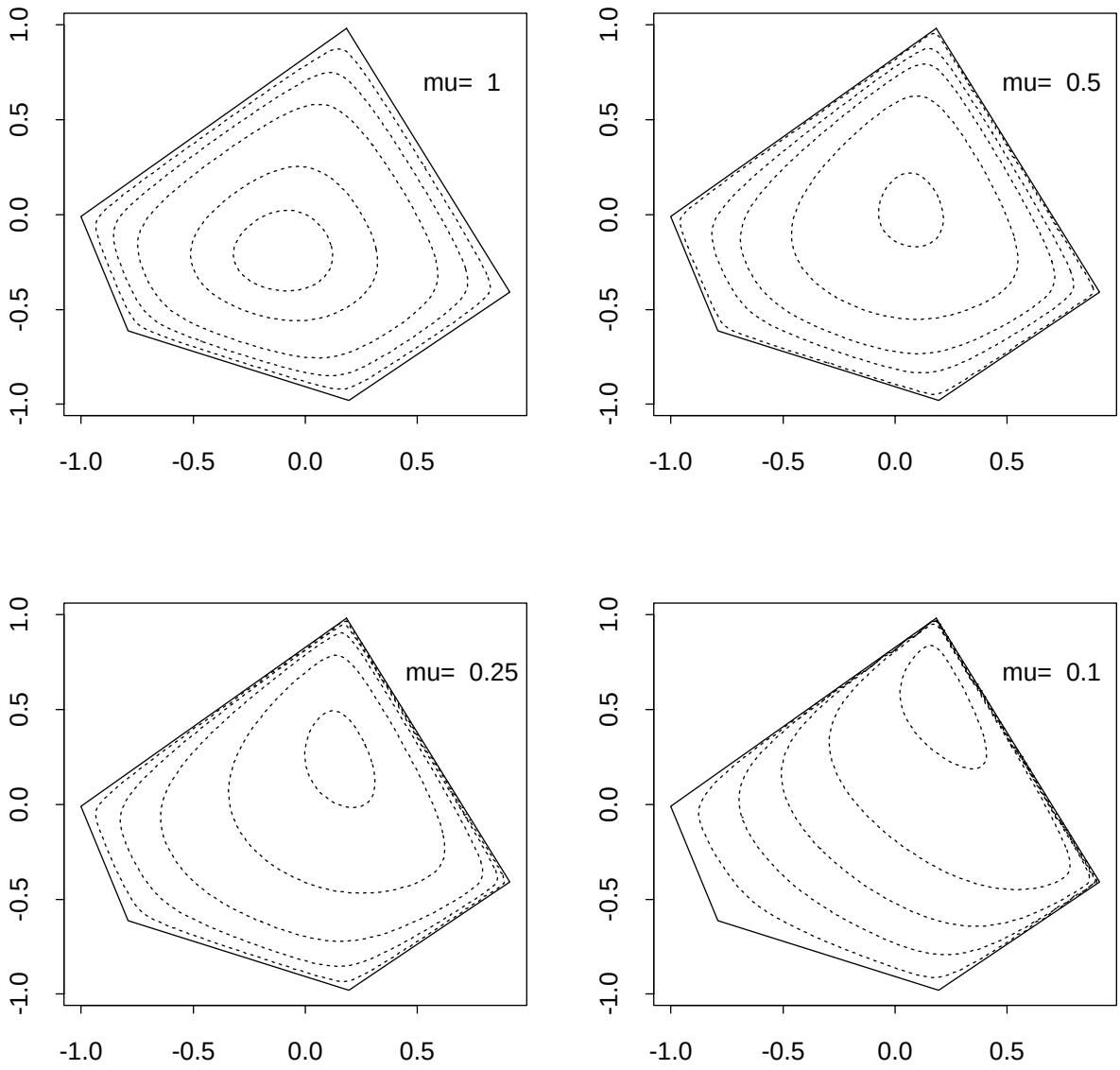


FIGURE 2.2. Contours of the Log Barrier Objective Function for the Simple Polygonal Linear Program: The figure illustrates four different contour plots of the log barrier objective function (2.8) corresponding to four different choices of μ. In the first panel, $\mu = 1$ and the contours are centered in the polygon. As μ is reduced the penalized objective function is less influenced by the penalty term and more strongly influenced by the linear component of the original LP formulation of the problem. Thus, for $\mu = .1$ we find that the unconstrained maximum of the log barrier function occurs quite close to the optimal vertex of the original LP. The locus of solutions to the log barrier problems for various μ's is called the central path, and is illustrated in Figure 2.1 by the solid curved line.

where e_n denotes an n-vector of ones. The simplex approach to solving this problem may be briefly described as follows. A p-element subset of $\mathcal{N} = \{1, 2, ..., n\}$ will be denoted by h, and $X(h), y(h)$ will denote the submatrix and subvector of X, y with the corresponding rows and elements identified by h. Recognizing that solutions to (2.1) may be characterized as planes which pass through precisely $p = \dim(b)$ observations, or as convex combinations of such "basic" solutions, we can begin with any such solution, which we may write in the primal formulation as,

$$b(h) = X(h)^{-1} y(h). \tag{2.5}$$

We may regard any such "basic" primal solution as an extreme point of the polyhedral, convex constraint set. In the dual formulation since the index set h identifies the active constraints of the primal problem, i.e. those observations for which both u_i and v_i are zero, $a(h)$ lies in the interior of the p dimension unit cube, and the complement of h corresponds to coordinates of a which lie on the boundary: if $u_i > 0$ then $a_i = 1$, while if $v_i > 0$ then $a_i = 0$. A natural algorithmic strategy is then to move to the adjacent vertex of the constraint set in the direction of steepest descent. This transition involves two stages: the first chooses a descent direction by considering the removal of each of the current basic observations and computing the gradient in the resulting direction, then having selected the direction of steepest descent and thus an observation to be removed from the currently active "basic" set, we must find the maximal step length in the chosen direction by searching over the remaining $n - p$ available observations for a new element to introduce into the "basic" set. Each of these transitions involves an elementary "simplex pivot" matrix operation to update the current basis. The iteration continues in this manner until no direction is found at which point the current $b(h)$ can be declared optimal.

To illustrate the shortcomings of the simplex method, or indeed of any strategy for solving linear programs which relies on an iterative path along the *exterior* of the constraint set, consider the problem depicted in Figure 2.1. We have a polygon whose vertices lie on the unit circle and our objective is to find a point in the polygon that maximizes the sum of its coordinates, that is, the point furthest north-east in the figure.

Since any point in the polygon can be represented as a convex weighting of the extreme points, the problem may be formulated as

$$\max\{e'u | X'd = u, \;\; e'd = 1, \;\; d \in \mathbb{R}^n_+\}, \tag{2.6}$$

where e denotes a (conformable) vector of ones, X is an $n \times 2$ matrix with rows representing the n vertices of the polygon and d is the vector of convex weights to be determined. Eliminating u we may rewrite (4.1) somewhat more simply as

$$\max\{s'd | e'd = 1, \;\; d \in \mathbb{R}^n_+\}, \tag{2.7}$$

where $s = Xe$. This is an extremely simple linear program which serves as a convenient geometric object for studying various approaches to solving such problems. Simplex is particularly simple in this context, because the constraint set *is* literally a simplex. If we begin at a random vertex, and move around the polygon until optimality is achieved, we pass through $\mathcal{O}(n)$ vertices in the process. This does not appear to be too onerous in Figure 2.1 where $n = 5$, but in ℓ_1 regression where the number of vertices is $\binom{n}{p}$ it can prove to be quite burdensome. Of course, a random initial vertex is rather naive, and one could do much better with an intelligent "Phase 1" approach that finds a *good* initial vertex. In effect, we can think of the "interior point" approach we will now describe as a class of methods to accomplish this, rendering unnecessary further travel around the outside of the polygon.

Although prior work in the Soviet literature offered theoretical support for the idea that linear programs could be solved in polynomial time, thus avoiding certain pathological behavior of simplex, the paper of Karmarker (1984) constituted a watershed in the numerical analysis of linear programming. It offered not only a cogent argument for the polynomiality of interior point methods of solving LP's, but also provided for the first time direct evidence that interior point methods were demonstrably faster than simplex in specific, large, practical problems.

It is an interesting irony, illustrating the spasmodic progress of science, that the most fruitful practical formulation of the interior point revolution of Karmarker (1984) can be traced back to a series of Oslo working papers by the economist Ragnar Frisch in the early 1950's. The basic idea of Frisch (1956) was to replace the linear inequality constraints of the LP, by what he called a log barrier, or potential, function. Thus, in our example, we may reformulate the problem as,

$$\max\{s'd + \mu \sum_{i=1}^{n} \log d_i | e'd = 1\} \tag{2.8}$$

where now the barrier term $\mu \sum \log d_i$ serves as a penalty which keeps us away from the boundary of the positive orthant. By judicious choice of a sequence $\mu \to 0$ we hope to converge to a solution of the original problem.

The salient virtue of the log barrier formulation is that, unlike the original formulation, it yields a differentiable objective function which is consequently attackable by Newton's method and inherits the quadratic convergence of Newton's method.

In Figure 2.2 we try to illustrate the log barrier approach by plotting 4 versions of the contours corresponding to the penalized objective function for four distinct values of the penalty parameter μ. In the first panel, with $\mu = 1$ we are strongly repelled from the boundary of the constraint set and the unconstrained maximum of the barrier function occurs near the center of the polygon. In the next panel, with μ reduced to $\frac{1}{2}$ the barrier penalty exerts a somewhat weaker effect and the contours indicate that the unconstrained maximum occurs somewhat closer to the upper vertex

of the polygon. This effect is further accentuated in the $\mu = \frac{1}{4}$ figure, and in the last figure with $\mu = \frac{1}{10}$ we find that the maximum occurs quite close to the vertex. The path connecting the maximum of the family of fixed-μ problems is generally called the central path. As emphasized by Gonzaga (1992) and others, this central path is a crucial construct for the interior point approach. Competing algorithms may be usefully evaluated on the basis of how well they are able to follow this path. Clearly, there is some tradeoff between staying close to the path and moving along the path, thus trying to reduce μ, iteration by iteration. Improving upon existing techniques for balancing these objectives is the subject of a vast outpouring of current research. Excellent introductions to the subject are provided in the survey paper of Margaret Wright (1992) and the recent monograph of Stephen Wright (1996).

In Figure 2.1 we have also illustrated the central path for our simple polygonal example. The solid curved line indicates the central path, the dotted line indicates the much superior first step taken by an affine scaling version of an interior point algorithm, while the dashed line indicates the first step of a primal dual version of the interior point method. The primal dual form of the algorithm requires somewhat more work per iteration, but takes fewer iterations and is somewhat more robust with respect to degeneracy and non-uniqueness of the solution. Formal computational complexity results indicate that for large problems primal dual implementations of interior point methods for solving ℓ_1 problems require $\mathcal{O}(np^3 \log^2 n)$ operations which is considerably better than the quadratic in n behavior of simplex, but still inferior to the $\mathcal{O}(np^2)$ behavior of least squares.

3. Preprocessing

But further gains are possible from careful preprocessing of ℓ_1 type problems. Preprocessing rests on an extremely simple idea: if, by preliminary estimation, or some other form of statistical necromancy, we could determine the signs of a significant group of observations, we could then combine observations with positive residuals into a single "globbed" observation, and similarly glob together the negative observations, so that the original problem,

$$\min \sum_{i=1}^{n} |y_i - x_i' b| \tag{3.1}$$

would be equivalent to,

$$\min \sum_{i \in N \setminus J_L \cup J_H}^{n} |y_i - x_i' b| + |y_L - x_L' b| + |y_H - x_H' b| \tag{3.2}$$

where $N = \{1, 2, ..., n\}$, $x_K = \sum_{i \in J_K} x_i$ for $K \in \{K, L\}$ and y_L and y_H can be chosen arbitrarily small and large respectively, to ensure that the corresponding residuals on the globbed observations remain negative and positive. In this process we have

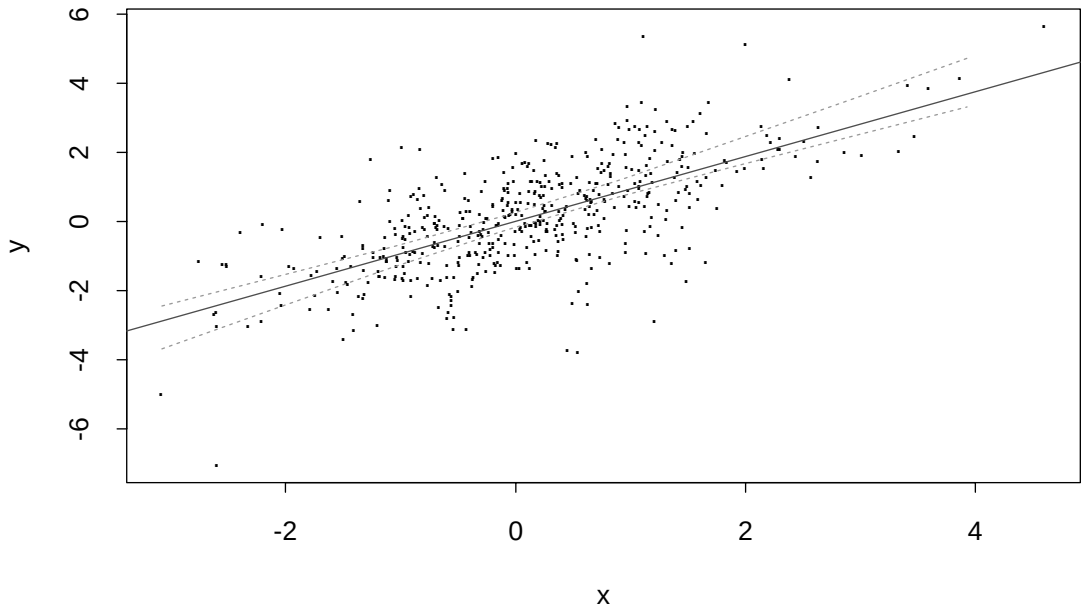


FIGURE 3.1. A Bivariate Example of Quantile Regression Preprocessing: The figure illustrates a bivariate scatter plot of 500 observations with y conditionally student t on 10 degrees of freedom. The curved dotted lines describe a confidence band for the response variable based on the median regression fit for a sub-sample of 126 observations. After globbing there are only 107 observations, including the two globbed observations. All the points outside the band are collapsed into this pair of pseudo-observations. The fit to the globbed sample is indicated by the solid line; since it falls inside the band we are assured that the globs are correct and that this solution is identical to a fit of the entire original sample.

reduced the problem of n original observations to $n - \sharp\{J_L, J_H\} + 2$ observations so if the cardinality of the J-sets is large we have gained substantially. Under plausible sampling assumptions we can, based on a preliminary subsample of m observations, make a prediction region for $\{x_i\beta : i = 1, 2, ..., n\}$ of width $\mathcal{O}(p/\sqrt{m})$, so assigning observations above this region to J_H and observations below this region to J_L, we would have $M = \mathcal{O}_p(np/\sqrt{m})$ observations falling inside the region. This is illustrated in Figure 3.1.

Minimizing the computational effort required to compute the preliminary fit based on m observations plus the effort required for the solution of the globbed problem (3.2) with M observations, we obtain $m^* = \mathcal{O}((np)^{2/3})$, which under our conjectured performance of the underlying interior point algorithm yields a complexity for the full problem of

$$C = \mathcal{O}_p(n^{2/3}p^3 \log^2 n) + \mathcal{O}(np^2), \tag{3.3}$$

where the first term comes from the solution of the two median regression problems of size $\mathcal{O}(n^{2/3})$ and the second term arises from the computation of the confidence band.

Further details are provided in Portnoy and Koenker(1997) and I will comment only briefly here on the important fact that any implementation of this preprocessing approach must verify that the solution to the globbed problem actually agrees with the predicted signs based on the confidence region. The simultaneous confidence region can be chosen to assure this with arbitrarily high probability, and the eventuality that we may need to repeat the cycle to remedy some inaccurately predicted signs introduces another multiplicative factor which does not affect the orders in probability in the complexity computation.

All of the foregoing discussion may be extended immediately from the median case to general quantile regression setting as introduced in Koenker and Bassett (1978). Quantile regression, by offering a means of estimating a complete family of conditional quantile functions in the familiar setting of linear regression, significantly extends the domain of applicability of ℓ_1 methods. In effect, it permits the researcher to examine the effect of specified covariates in particular ranges of the conditional distribution of the response variable, and thus to distinguish differential effects in the tails, or on dispersion, from the classical regression effect characterized by a simple location shift.

The crucial consequence of the formal complexity theory and the extensive concomitant empirical testing of our implementation of the algorithm is that the computational effort required for quantile regression can be made comparable with the effort required for least squares over the full range of currently plausible problem dimensions. In the final empirical example of Portnoy and Koenker (1997), we compare timings for a typical large econometric application of quantile regression with $n = 113,547$ and $p = 6$. With the new algorithm, quantile regression estimates take about 10 seconds on a Sparc-Ultra, comparable to the least squares time of 8 seconds. Interior point methods applied to the full problem before preprocessing requires about a minute for these problems. Simplex solution of the same quantile regression problems requires approximately an hour on the same machine.

4. Future Prospects

There are many open questions posed by the rapid development of computational methods for ℓ_1 methods and quantile regression more generally. But we are, I believe,

on the verge of fully vindicating Edgeworth's old claim that the "plural median" is less laborious, as well as more robust, than its least squares competitor. Laplace's Tortoise has a real chance against Hare "Gauss", and it is to be hoped that this will provide a continued impetus for the further development of these methods in statistics.

University of Illinois, Urbana-Champaign

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